

R487: Outer Bridges, Lane Start

TFPT RH programme

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Claim boundary. This round types known classical obligations, proves two logical wrappers, and splits one analytic gap. Four named classical bricks remain. There is no Riemann-hypothesis claim.

1 Inventory and work plan

Bridge	Missing pieces	Mathlib position	Estimate
Grid density	Dyadic step approximation of a compactly supported L^2 autocorrelation witness; support control; pointwise and L^1 convergence; continuity of arch, finite prime, and polar channels.	Bochner/interval integrals, topology and dominated convergence exist. No dedicated autocorrelation-density theorem.	6–10 rounds, about 3–5 weeks.
Explicit formula	Match all constants and transforms in the prime, archimedean and polar channels; construct the zero spectral side, local finiteness, multiplicity and Guinand–Weil equality.	<code>riemannZeta</code> , analytic continuation and functional equation exist. No explicit-formula theorem, zero enumeration or multiplicity API. Mellin transform/inversion and Mellin-to-Dirichlet theorems exist.	16–28 rounds, about 8–14 weeks.
Weil criterion	For every off-critical nontrivial zero, construct a fixed-support autocorrelation with negative standard form; handle conjugate and functional-equation symmetries.	<code>RiemannHypothesis</code> is exactly a quantified zero statement, so the last interface is elementary. The separating-test theorem is absent.	8–14 rounds, about 4–7 weeks after the spectral API.

These are effort estimates for formalization, not mathematical novelty claims. The explicit-formula bridge dominates the lane.

2 Dense bridge reduction

`FullWeilTest` now records a continuous even function supported in $[-R, R]$, $R \geq 0$, together with an explicit real autocorrelation witness

$$g(u) = \int_{\mathbb{R}} h(t)h(t+u) dt.$$

`FixedSupportGridApproximation` requires support preservation, pointwise convergence on the support, and $L^1([-R, R])$ convergence. The open construction is exactly `fullWeil_fixedSupport_grid_density`.

The form is decomposed visibly into archimedean, von-Mangoldt and polar channels. Their continuity is the second named brick `fullWeil_channel_continuity`. Once these two bricks are supplied, the following are proved in Lean:

- `fullWeilForm_tendsto_of_channels`;
- `fullWeilForm_nonneg_of_tendsto`;
- `grid_dense_extension_of_fixedSupport`;
- the public wrapper `grid_dense_extension`.

The positivity step uses closedness of $[0, \infty)$, through `ge_of_tendsto`; no mesh-PSD transport or infinite-window research hypothesis is introduced.

3 Zeta-interface design

Mathlib 4.29.1 defines

$$\text{RiemannHypothesis} = \{\rho : \zeta(\rho) = 0, \rho \text{ nontrivial}, \rho \neq 1 \implies \Re \rho = \tfrac{1}{2}\}.$$

There is no `ZetaZero` structure or enumeration. R487 therefore uses a direct predicate built from `riemannZetas=0`.

The spectral target is the named proposition `GuinandWeilExplicitFormula`: the arithmetic three-channel form equals the multiplicity-weighted zero spectral form, once that spectral API has been built. The criterion target is `FullWeilSeparatesOffCriticalZeros`: each off-critical nontrivial zero admits an admissible autocorrelation with negative standard form. The theorem `standard_weil_criterion_to_mathlib_rh_of_separation` proves the remaining logical implication by contradiction, and the public `standard_weil_criterion_to_mathlib_rh` is its wrapper.

References for the future theorem are Weil, *Sur les formules explicites de la théorie des nombres premiers* (1952), and Bombieri, *The Riemann Hypothesis* (Clay Mathematics Institute, 2000).

4 Census and kill table

The old three external sorries become four: density was honestly split into construction and continuity; explicit-formula normalization remains one; the old opaque final criterion is replaced by one precise off-critical separation theorem. Repository census: $8 \rightarrow 9$.

Candidate	Status	Reason
Mellin infrastructure closes the bridge alone	killed	Transform, inversion and Mellin-to-Dirichlet results exist, but no Guinand–Weil formula or zero spectral sum is supplied.
Mathlib already has a zeta-zero API	killed	Zeros are represented only by the equation <code>riemannZetas=0</code> .
Positivity survives the limit	proved	<code>ge_of_tendsto</code> closes the order-topological step.
Standard positivity implies Mathlib RH	reduced	The interface is proved from the single named separation theorem.
R487 proves RH	killed	Four external sorries and internal hypotheses remain.